

PitchIQ

MLB Pitcher Scouting Dashboard

AQX Sports Analytics Hackathon | Technical Methodology & Benchmark Reference

Live Demo: <https://pitchiq-aqx.streamlit.app/>
GitHub: <https://github.com/trinik15/pitchiq>

1. Project Overview

PitchIQ is an interactive MLB pitcher scouting dashboard built on live MLB Statcast data. It gives coaches, scouts, and front-office analysts pitch-by-pitch intelligence - including per-pitch letter grades benchmarked against 2024 MLB averages, platoon-split filtering, count-tendency heatmaps, and year-over-year comparisons - in a clean, immediately usable interface.

Core design principle: surface the same information a pitching coach builds manually from raw Statcast exports, but deliver it in under 5 seconds with no SQL, no spreadsheets, and no analytics background required.

2. Statistical Formulas & Definitions

Whiff%

Swinging strikes / Total swings x 100
Swings = swinging_strike + swinging_strike_blocked + foul + foul_tip + hit_into_play (all variants) + missed_bunt + foul_bunt
Whiffs = swinging_strike + swinging_strike_blocked only
Industry-standard definition (FanGraphs / Baseball Savant).

CSW%

(Called strikes + Swinging strikes) / Total pitches x 100
Captures both the pitcher's ability to get strike calls AND generate swings-and-misses.
Higher CSW% = more pitch-level dominance per pitch thrown.

Chase%

Out-of-zone swings / Total out-of-zone pitches x 100
Out-of-zone defined as Statcast zone codes 11-14 (ball zones outside strike zone).
Measures how effectively the pitcher induces batters to chase balls.

Weighted CSW% (header tile)

Sum(pitch_type_CSW% x pitch_type_usage_fraction) across all pitch types.
Single number representing overall strikeout-generating efficiency.

3. Grading System - True Z-Score vs. 2024 MLB Baselines

Each pitch is graded A+ through F using a z-score calculated against 2024 MLB averages for that specific pitch type (Baseball Savant, min 200 pitches per pitcher-season). PitchIQ uses pitch-type-specific standard deviations (sigma), not a universal window.

Z-score: $z = (\text{pitcher_value} - \text{MLB_avg_for_pitch_type}) / \text{sigma_for_pitch_type}$

Grade thresholds:

Grade	Z-Score	Interpretation
A+	$z \geq +1.5$	Elite - top ~7% of MLB
A	$z \geq +0.75$	Above average - top ~23%
B	$z \geq +0.20$	Slightly above average
C	± 0.20	League average range
D	$z \leq -0.75$	Below average
F	$z \leq -1.5$	Well below average - bottom ~7%

Each pitch receives four independent grades: Velocity, Spin Rate, Whiff%, CSW% (plus Chase% when zone data is available). The overall badge grade averages these dimension scores and converts back to a letter.

4. MLB 2024 Benchmarks by Pitch Type (Baseball Savant, min 200 pitches)

All grading is pitch-type-specific. The table lists mean (μ) and standard deviation (σ) used for z-score computation. Sigma represents the inter-pitcher distribution for 2024 qualified starters - how much pitchers vary from each other for each metric.

Pitch Type	Velo μ	Velo σ	Spin μ	Spin σ	Whiff% μ	Whiff% σ	Chase%	Chase% σ	CSW% μ	CSW% σ
4-Seam FB (FF)	94.1	2.5	2262	220	23.8%	8.0%	28.1%	7.5%	28.5%	5.8%
Sinker (SI)	93.4	2.3	2183	195	16.1%	5.5%	30.2%	7.0%	25.4%	4.8%
Cutter (FC)	89.2	2.8	2382	225	24.3%	9.0%	30.8%	8.2%	30.2%	6.5%
Slider (SL)	84.2	3.5	2405	275	33.2%	10.5%	35.1%	9.0%	33.4%	7.5%
Sweeper (ST)	82.5	3.2	2395	265	37.8%	11.0%	38.2%	10.0%	34.1%	7.8%
Curveball (CU)	78.5	3.8	2526	310	28.4%	9.5%	34.5%	9.5%	31.2%	7.0%
Knuckle Curve	76.8	3.5	2598	290	26.2%	9.0%	32.1%	8.5%	30.1%	6.8%
Changeup (CH)	85.2	3.0	1790	245	34.5%	10.5%	38.8%	10.0%	33.1%	7.8%
Splitter (FS)	85.8	3.2	1675	220	37.8%	11.0%	40.2%	10.5%	34.2%	8.0%

Analytical insight: Offspeed and breaking balls have higher sigma for Whiff% and CSW% because pitcher quality varies more widely for secondary pitches. A 45% Whiff% sweeper grades A+ ($z \sim +0.7$), while the same 45% on a splitter grades A ($z \sim +0.6$) - because splitter specialists are naturally exceptional whiff generators. Pitch-type-specific sigma is essential for fair cross-pitcher comparison.

5. Count Tendency Analysis & Auto-Generated Insight Algorithm

The Count Tendencies tab shows pitch selection across all 12 counts (0-0 through 3-2), computed live from raw pitch logs. The auto-generated insight callout identifies the most exploitable count/pitch combination:

1. Compute season-average Whiff% for each pitch type.
2. For each of 12 counts (≥ 15 pitches), compute per-pitch Whiff% (min 5 pitches per cell).
3. Calculate Delta = count Whiff% - season-average Whiff% for that pitch type.
4. Surface the (count, pitch) with the largest positive Delta ≥ 10 percentage points.
5. Display: 'In [count] counts, [pitcher]'s [pitch] generates [X]% Whiff - [D] pts above season avg.'

Minimum sample gates: 20 pitches in the count situation; 10 pitches per count/pitch cell.

This gives scouts actionable information with no analogous automated feature in free MLB tools.

6. Platoon Split Methodology

The Batter Handedness toggle (All / vs LHB / vs RHB) filters the Statcast pitch log by the 'stand' column, then recomputes ALL metrics, grades, and charts from scratch. Every number - z-scores, letter grades, heatmaps, insight callout - reflects the selected handedness split.

This is analytically critical: a pitcher's sweeper may grade A vs RHB but C vs LHB. A scout preparing a left-handed hitter needs split-specific grades, not aggregate numbers. No free public tool provides this in an integrated, all-metrics view.

7. Recommendation Engine Methodology

The Scouting Recommendation card on the Summary tab surfaces the single most actionable platoon exploit supported by the data. The algorithm:

1. For each pitch type thrown by the pitcher, compute Whiff% separately against LHB and RHB.
2. Apply a minimum sample gate: both sides must have at least 80 pitches. If either side falls below 80, that pitch type is excluded from consideration.
3. Compute the absolute delta: $|\text{Whiff\%}_{\text{LHB}} - \text{Whiff\%}_{\text{RHB}}|$.
4. Apply a minimum delta threshold: the delta must be ≥ 15 percentage points.
5. If any pitch type passes both gates, surface the one with the largest delta as the recommendation, including exact sample sizes and the direction of the split.
6. If no pitch type passes both gates, display an explicit 'insufficient data' message rather than fabricating a recommendation.

Example output for Paul Skenes 2025:

"Skenes's Slider generates 50% Whiff vs LHB vs only 24% vs RHB - a 26pp split (n=65 vs LHB, n=103 vs RHB). Consider increasing usage in this matchup."

Honesty gate design principle: the recommendation engine either produces a specific, statistically-backed claim with exact sample sizes and delta, or it says nothing. No middle ground, no vague language, no low-sample extrapolation.

8. Contact Quality & xwOBA Methodology

The Scouting Summary tab includes a Contact Quality section that measures what happens when batters make contact, not just swing-and-miss outcomes.

For each pitch type with at least 20 balls in play (BIP):

- GB%, LD%, FB%, PU% from Statcast bb_type classification
- avg_xwoba from estimated_woba_using_speedangle (lower = weaker contact allowed)
- avg_exit_velo from launch_speed

Rows are sorted ascending by avg_xwoba so the pitch inducing the weakest contact surfaces first. MLB average xwOBA on contact is approximately 0.320.

This complements Whiff% and CSW% grading: a pitch with modest whiff but elite contact suppression (e.g., Skenes

Changeup at 0.236 xwOBA) is correctly valued as a different weapon than a swing-and-miss pitch.

9. Matchup Prediction Methodology

The Matchup tab answers: given a specific count and batter handedness, what pitch does this pitcher historically throw, and how effective is each option?

Algorithm:

- 1. Filter pitch log to selected count_label and batter stand (L/R).
- 2. Require minimum 15 pitches in the filtered situation; otherwise display 'insufficient data'.
- 3. Compute usage %, Whiff%, and CSW% per pitch type in that situation.
- 4. Generate situational insight: if dominant pitch >= 50% usage, flag as 'hitter should be sitting on this pitch'; if >= 35%, flag as primary pitch; otherwise report even distribution.

Use case: advance scouts preparing a hitter for a 1-2 count vs a left-handed batter see the pitcher's historical mix and effectiveness before stepping in the box - without manual Statcast exports.

10. Data Pipeline

Source:	MLB Statcast via pybaseball (public, no API key required)
Coverage:	2024-2026 MLB regular and postseason pitch logs
Granularity:	Individual pitch rows (~150 features: velo, spin, break, location, outcome)
Demo data:	7 pitchers bundled as .csv.gz: Paul Skenes (2024/2025), Gerrit Cole (2024/2025), Shohei Ohtani (2025), Zack Wheeler (2024/2025), Emmanuel Clase (2024/2025), Sandy Alcantara (2025), Blake Snell (2024/2025). Loads in under 200ms, no network required.
Live fetch:	Any qualified MLB pitcher via playerid_lookup + statcast_pitcher (~60s)

Statcast is the gold standard for pitch tracking, measured by Hawk-Eye cameras at every MLB stadium. Each pitch record includes: release velocity, spin rate, pfx_x/pfx_z (induced break in feet), release coordinates, plate crossing location, count, zone code, and outcome.

11. Technology Stack

UI Framework:	Streamlit 1.45.1
Data Fetching:	pybaseball 2.2.7 (MLB Statcast wrapper)
Charts:	Plotly 5.22 (interactive, dark-themed)
Data Processing:	pandas 2.2.3 / NumPy 1.26.4
Hosting:	Streamlit Community Cloud (free tier, public URL, no account required)
Language:	Python 3.11

12. Competitive Differentiation vs. Free Tools

Feature	PitchIQ	Savant	FanGraphs
Pitch-type-specific grading (A+ to F)	YES	No	No
True z-score benchmarks vs. peer pitch types	YES	No	No
Platoon-split grades (vs LHB / vs RHB)	YES	Partial	Partial
Count-specific zone heatmaps	YES	Manual	No

Auto-generated insight callout	YES	No	No
Actionable platoon recommendation w/ sample gate	YES	No	No
Contact quality / xwOBA by pitch type	YES	Manual	No
Situational matchup pitch-mix card	YES	Manual	No
All metrics in one view, one pitcher	YES	Multi-pg	Multi-pg
Year-over-year velocity/usage trend	YES	Manual	Yes
CSV export of pitch summary	YES	Yes	Yes

Strongest differentiator: pitch-type-specific z-score grading with live platoon splits, combined with a recommendation engine that either produces a specific, statistically-backed actionable insight or explicitly withholds one. No free tool shows a scout - in one view - exactly how a pitcher's curveball grades against its peer curveballs, separately for LHB and RHB, with a direct recommendation on which platoon to exploit. That is the core gap PitchIQ addresses.

13. Challenges & Learnings

1. Data volume: A full starter season is 3,000+ rows x 150 columns. Bundling 7 pitchers (Skenes, Cole, Ohtani, Wheeler, Clase, Alcantara, Snell) as compressed CSVs reduced first-load from 90s to under 200ms for demo pitchers.
2. Grading calibration: With a universal window, Skenes' splitter (42.3% Whiff%) graded only C because a universal window of 12pp treated it as near-average. With pitch-type-specific sigma (FS sigma = 11.0%), it correctly grades A+ - because that's genuinely elite for a splitter. The real sigma was essential for accurate grading.
3. Platoon filter integrity: Applying the handedness filter at the data level (before all computations) - not at the chart level - ensures all z-scores, benchmarks, and insights are internally consistent within the selected split.
4. Claims accuracy: Minimum sample gates (80 pitches per platoon side, 20 pitches per count cell, 10 transitions per sequencing pair) prevent the system from surfacing insights built on noise. The honesty gate became a core design principle: if data does not support a claim, say so explicitly.